

Canvas LMS Auto-Grading Tool Integration and Architecture Planning - AI summary

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Conversation summary

The team discussed transitioning physical lab manuals and exams into a digital, auto-graded format integrated with Canvas LMS. They evaluated rule-based Python scripts versus AI semantic matching, addressed FERPA compliance and LTI integration challenges, and outlined a custom API and local testing workflow to automate grading while maintaining visual document standards.

- Digital Worksheet Transition and Grading Methodology
 - Context: The team is discussing the end-to-end transition of physical lab manuals, worksheets, and exams into a digital format integrated with Learning Management Systems (LMS) like Canvas and Blackboard. This includes exploring automated grading methods (both rule-based and AI-driven), addressing technical integration challenges via LTI and custom APIs, ensuring FERPA compliance and data privacy, and optimizing the user interface and overall workflow for both students and instructors.
 - Digital PDF and HTML Format Evaluation: Students could use visually appealing PDFs on laptops to input experimental data, maintaining physical printout aesthetics. Claude's HTML proposal was rejected as visually unappealing (resembling "boxes"), preferring PDFs replicating traditional printed papers.
 - Acceptable Answer Ranges and Partial Credit: Instead of single fixed values, specific acceptable ranges will determine correctness. Partial credit cannot easily be awarded for Python code or other inputs; answers must fall strictly within defined ranges.
 - Rule-Based vs. AI Grading Decision: The consensus rejected AI (like UTD's "Comet AI") for grading, as exact answers are known, making a rule-based system sufficient. AI is not recommended due to grading variance risks.
 - Python Similarity Thresholding and Extraction: A 90% similarity threshold will automatically group/score answers. A custom Python script will parse downloaded student files in a single folder, compiling grades into Excel.

- Non-AI vs. AI Semantic Matching Approaches: A non-AI rule-based approach is complex, requiring extensive coding for answer variations. An AI approach would use semantic matching against benchmark keys, though physics relies on factual, structured answers.
- Flagging, Deductions, and Tabular Data: Auto-grading text directly into tabular data containing specific ranges is straightforward. The system must flag undesired student responses and automatically deduct points.
- Qualitative Answer Benchmarking: To evaluate and benchmark qualitative, word-based answers rather than numerical data, a reference benchmark exists with 19 specific items available.
- Two-Stage Implementation Proposal: A suggestion was made to divide the project into two distinct stages to optimize the initial phase of the auto-grading process before transitioning to the next.
- Canvas LMS Integration and Custom LTI Tool Development
 - Context: The team is discussing the technical integration of their custom grading tool with Canvas LMS, exploring LTI standards, API capabilities, developer keys, and the workflow required to sync grades directly to the gradebook.
 - Custom LTI Tool Registration and Python Support: The new tool is built specifically for Canvas using LTI standards. Integration requires creating a developer key, enabling the tool to run a Python script triggered upon submission.
 - Canvas API Feasibility and Knowledge Gap: Necessary Canvas APIs exist, but the team acknowledged a knowledge gap regarding Canvas's internal workings, agreeing to research documentation. Canvas LMS source code is accessible via Git.
 - Proposed Custom Auto-Grading Workflow: The tool will utilize Canvas's Assignment Grading Service API. Upon submission, assignments route through the custom system to generate grades, initially visible only to instructors.
 - Grade Visibility and Gradebook Sync: A key constraint is that students must not see grades directly in processed files or the gradebook immediately, allowing instructors to review and tweak scores before release.
 - Rejection of PDF Upload Workflow: The team rejected ChatGPT's proposed workflow where students work on a PDF and upload it. Instead, students should edit documents directly in Canvas and click submit.
 - Local Testing Environment and Lattice Reference: The team plans to set up a local

Canvas LMS instance to verify functionality. A system/paper called "Lattice" was mentioned as a reference point.

- **Data Privacy, FERPA Compliance, and University Approval**

- **Context:** The team addresses the administrative, legal, and security hurdles of deploying their custom tool, focusing on FERPA compliance, data leakage risks, and university approval processes.
- **FERPA Compliance and Auditing:** The integration must comply with FERPA policies. The proposed system will run a quick audit to verify compliance, requiring research into Canvas's alignment with FERPA.
- **Data Leakage and External AI Risks:** Protecting student information restricts external AI tools to avoid exposing sensitive details like names. If no identifying details are sent, privacy risks are mitigated.
- **University Approval Constraint:** A major point of friction is whether the University of Texas at Dallas (UTD) will officially approve the integration and use of this custom-developed tool.

- **Current Grading Workflows and Platform Limitations**

- **Context:** The team reviews existing manual grading processes, file formats, and limitations of current platforms like Blackboard and Turnitin to highlight the need for automation.
- **Blackboard Platform Manual Grading and Limitations:** The current system is Blackboard, not Canvas. Students upload documents, and teachers must manually view submissions and input scores. The interface limits submissions to a single document.
- **Turnitin LTI System Limitations:** Currently, students upload to Turnitin, requiring instructors to manually input grades. It lacks AI or automation to feed scores directly into the gradebook.
- **Lab Assignment File Formats and Layout Issues:** Assignments are written in Word and distributed as PDFs. The original design has grading boxes on the bottom right, making manual calculation and tracking difficult.
- **PDF Usability and Acrobat Licensing:** The institution licenses Acrobat, allowing interactive PDFs. Users sometimes struggle with extracting text, but PDFs are generally considered easier to handle and support value extraction.

- Miscellaneous Technical Discussions, Tool Demos, and Meeting Logistics
 - Context: This section aggregates secondary discussions, including AI verification issues, school ID card costs, hosting constraints, autonomous agent logging tools, and meeting logistics.
 - ChatGPT Verification and Excel Hallucinations: ChatGPT exhibited hallucinations during verification, generating random student names and numerical values in an Excel file instead of prompting for actual inputs.
 - School ID Card Implementation Cost: Implementing a specific card/system at UTD is estimated to cost \$1-\$2 for 100 people. UTD does not currently use this system but may acquire it.
 - Software Hosting and Personnel Cost Constraints: Running the software is highly doable and easy, but there is a strict constraint not to pay for Teaching Assistants (TAs) or personnel to manage it.
 - Autonomous Agent Event Logging and Replay Tool: A developer built a standalone tool to log and replay actions executed by autonomous agents. The tool is fully built and spun out from a larger project.
 - Audit Framework for AI Agents: A framework designed for auditing AI agents integrates with agents to sign every action with a signature. It includes a development kit and runs in an active product.
 - Meeting Interruption for Calendar Conflict: A participant left the meeting briefly for approximately 10 minutes due to an unscheduled calendar conflict, which was acknowledged by the other participants.
 - Yeah: 无具体背景

Action items

- Email Lattice research papers to the professor
- Research Canvas LMS documentation and codebase
- Research FERPA compliance and data privacy for Canvas integration
- Set up local Canvas LMS testing environment